

Fear Is a Bad Gambler: Designing for Cognitive Co-Creation in AI-Mediated Social Exposure

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Abstract

In this paper, we present an interactive system based on the Prediction–Exposure–Rewriting (P–E–R) framework to support self-guided exposure for Social Anxiety Disorder (SAD). Grounded in inhibitory learning theory and cognitive behavioral principles, the system leverages a logic-driven Large Language Model (LLM) as a co-creative partner to help users collaboratively formalize vague anxieties into quantifiable predictions, which are then explored through dynamically generated social simulations designed to maintain realistic yet safe interaction dynamics. An independent auditor agent continuously monitors scenario severity to prevent distress escalation, while an externalization architecture materializes the gap between user predictions and observed outcomes to prompt active belief revision. Through voice-based interaction, the system supports 1) probabilistic fear articulation, 2) structured social exposure with adaptive co-regulation, and 3) guided post-interaction reflection. In a pilot study, we found that explicitly structuring prediction, experience, and reinterpretation helps users recalibrate threat expectations and engage more confidently with anxiety-provoking social situations.

CCS Concepts

• **Human-centered computing** → **Systems and tools for interaction design**; *HCI design and evaluation methods*.

Keywords

Social Anxiety Disorder, Exposure Therapy, Generative AI, Co-creation

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1 Introduction

Social Anxiety Disorder (SAD) is characterized by a fear of negative evaluation and persistent avoidance of social situations, often resulting in psychological impairment. A core cognitive feature of SAD is catastrophic thinking, which involves the systematic overestimation of the likelihood and severity of negative social

outcomes [7, 13, 16]. Cognitive models suggest that these exaggerated threat predictions maintain anxiety by reinforcing negative self-beliefs, particularly when corrective social evidence is ignored [6]. Clinical research supports exposure therapy, which involves repeated confrontation with feared situations, as a primary treatment for testing these predictions [9, 12]. While exposure may be conducted in real-world or virtual settings, independent implementation presents challenges regarding the articulation of specific feared outcomes and the management of distress [4, 10]. As Kendall et al. (2019) noted, while outcomes help map out evidence in an uncertain terrain, their meaning remains relative to the individual's pre-existing cognitive biases [14]. Without structured guidance, exposure attempts may not lead to the intended updates in threat predictions.

To support this process, we present an interactive system based on the Prediction–Exposure–Rewriting (P–E–R) framework. The system structures exposure as a behavioral experiment by enabling users to externalize probabilistic predictions, validating these predictions within dynamically generated social simulations, and facilitating post-interaction reflection. By integrating Large Language Model (LLM) scenario generation with multimodal monitoring, the system aims to support cognitive change and maintain safety during self-guided exposure.

2 Related Work

Digital interventions for Social Anxiety Disorder (SAD) aim to reduce avoidance by providing self-guided therapeutic tools and alternative forms of exposure [11, 17, 28, 29]. Grounded in cognitive models that attribute SAD to maladaptive threat appraisals and catastrophic expectations of negative social evaluation [24], these digital formats lower practical barriers to treatment and support repeated engagement with anxiety-provoking situations, making them a common component of contemporary anxiety interventions [15]. Recent advances in generative artificial intelligence further expand the design space for digital mental health systems [26]. Large Language Models (LLMs), together with multimodal sensing, enable the generation of socially relevant scenarios and the tracking of user signals during interaction, allowing AI-mediated systems to approximate social uncertainty in controlled settings [18, 21, 23].

Within HCI, systems that externalize internal states to support reflection and behavior change have gained increasing attention [30, 31]. However, existing approaches reveal two persistent gaps. First, exposure-based systems often reduce intervention to repeated confrontation, emphasizing habituation while neglecting expectancy-violation mechanisms central to inhibitory learning [9, 14]. Second, although AI-generated interactions are increasingly



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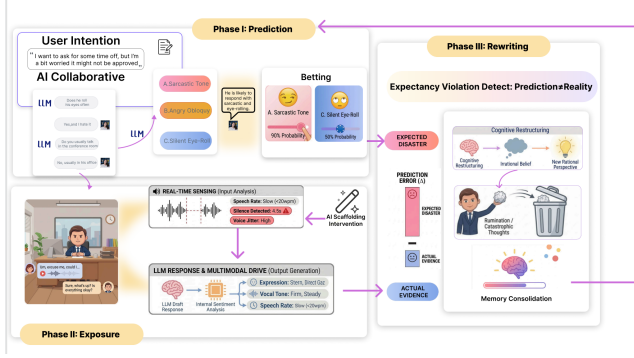


Figure 1: System overview of the Prediction-Exposure-Rewriting (P-E-R) framework

rich, they are typically unstructured, positioning AI as a conversational partner rather than a mechanism for testing and revising maladaptive predictions [1]. To address these gaps, we adopt a cognitive co-creation perspective [20], drawing on augmented cognition to operationalize exposure as a structured behavioral experiment. This perspective motivates a shift toward designing systems that link anticipation, experience, and reinterpretation.

3 System Design

We operationalize the P-E-R framework into a closed-loop interactive system (Figure 1). The architecture integrates LLMs with multimodal sensing to facilitate a structured cognitive intervention.

3.1 User Research and Design Goals

Through semi-structured interviews with three licensed therapists and eight individuals with Social Anxiety Disorder (SAD), we identified three key limitations in independent exposure practices. Interview transcripts were thematically analyzed to extract recurring patterns across clinical and lived experiences. Based on these findings, we derived three design goals (DGs) to guide the system design. First, participants struggled to define fears concretely. Their descriptions reflected a tendency toward catastrophic thinking, in which negative social outcomes were systematically overestimated. This motivated our first design goal **DG1: Co-construct Concrete Anxiety Models by visually externalizing abstract anxieties into inspectable representations**. Second, self-guided exposure posed safety risks without real-time monitoring or external support. Unregulated attempts could trigger overwhelming distress, panic, or avoidance reinforcement, and therapists emphasized paced, co-regulated exposure as critical for individuals with high baseline anxiety. These insights informed **DG2: Maintain Safety through Adaptive Co-regulation by monitoring user distress and providing dynamic, non-intrusive support**. Third, even when social interactions unfolded without feared outcomes, participants' maladaptive beliefs often persisted due to cognitive distortions and negative post-event reinterpretation. This gap between exposure experience and belief updating motivated **DG3: Materialize Cognitive Conflicts by externalizing discrepancies between expected and simulated outcomes to support reflection and belief revision**.

Table 1: System Framework: Mapping Therapeutic Goals to Computational Modules

Phase	Therapeutic Challenge	Cognitive Mechanism	Computational Module	Materialized Output
P1	Ambiguous fear	Testable hypotheses (CBT)	Probabilistic modeling engine	Digital bet (P_{pre})
P2	Safety escalation	Expectancy violation (Inhibitory learning)	Constrained social generation + Auditor (O_{bet})	Normative reaction
P3	Belief persistence	Prediction-error-driven updating	Externalization architecture (Evidence Court)	Update artifact (Δ)

3.2 Mechanism Design: Theoretical Translation

Our system operationalizes the P-E-R framework by establishing a direct isomorphism between therapeutic protocols and computational modules.

3.2.1 Phase I — Anxiety Formalization (Probabilistic Modeling Engine). In traditional Cognitive Behavioral Therapy (CBT) [3], anxious thoughts are treated as testable hypotheses instead of objective facts. Our system facilitates this shift by functioning as a probabilistic modeling engine: when a user inputs a vague apprehension, the system scaffolds the clarification of a specific, falsifiable outcome. This process converts an internal emotion into a *Digital Bet* with a measurable probability, allowing users to externalize their fear as a discrete object for cognitive restructuring.

3.2.2 Phase II — Inhibitory Learning (Constrained Generation Pipeline). To facilitate fear unlearning, the system provides a controlled environment in which catastrophic expectations are systematically challenged. We implement a constrained generation pipeline in which AI agents adhere to a *Normative Social Baseline*, ensuring reactions remain socially typical and neutral. By maintaining behavioral stability, the system creates a 'fair test' for the user's predictions. Rather than providing an artificial safety net, this constraint approximates ecological reality by anchoring the AI's responses in the statistically typical, neutral-to-mildly-negative outcomes of real-world social encounters. When the AI fails to exhibit the anticipated hostility learning [2, 8].

3.2.3 Phase III — Belief Updating (Externalization Architecture). The final stage ensures that discrepancies between anticipation and reality are explicitly recognized and integrated. The system employs an externalization architecture to visually contrast the initial *Digital Bet* with the audited events of the simulation. Rather than presenting a passive summary, the interface materializes the "Delta" (prediction error) as an interactive visual object. This enforced confrontation encourages users to update their internal models and rewrite maladaptive beliefs before the session concludes.

3.3 User Journey: The Interaction Experience

The P-E-R framework comes to life through Alice, who feels anxious about asking her boss for time off. She starts at the **Betting Interface**. Instead of letting her fear stay vague, the system helps her turn it into a specific prediction: "My boss will yell at me." Alice then drags a **90% Probability Chip** onto a digital *Fear Card* (the visual representation of the Digital Bet). This simple act turns her overwhelming emotion into a clear "bet" that she can objectively test (Figure 2).

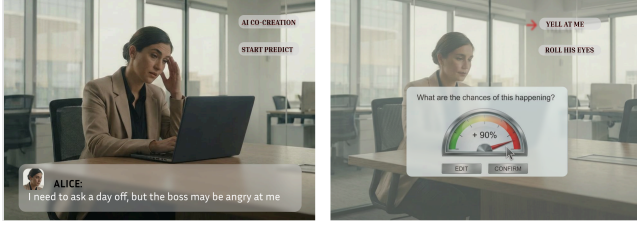


Figure 2: Prediction stage interface.

Next, Alice enters the **Social Simulation** to practice the conversation. As she speaks, the system detects her stress through her hesitant speech. To help her stay calm, a gentle breathing cue pulses on the edge of the screen. This guides her breathing without distracting her from the talk. Even if Alice stutters, the virtual manager stays professional and neutral. This calm response directly challenges Alice’s expectation that she would be met with hostility.

The journey ends in the **Evidence Court**. Here, Alice’s original 90% prediction is placed side-by-side with the actual record of the manager’s neutral reaction. Seeing this gap helps her realize her fear did not match reality. Alice then performs one final action: she drags the disproven Fear Card away and deletes it. This physical movement signals that the old belief has been tested and no longer controls her expectations.

4 System Implementation

To realize the P-E-R mechanism, we engineered a scalable system architecture that integrates Large Language Models (LLMs) for generative capabilities and a multimodal pipeline for real-time sensing. The system is built on a React frontend and a Python Flask backend, utilizing WebSocket for low-latency communication [19].

4.1 Comprehensive Voice Analysis Pipeline

Once the data reaches the backend, the system divides the audio into two parallel processing tracks to capture a comprehensive representation of the user’s speech: (1) Semantic Track, in which the Whisper Large-v3 model transcribes the conversation into text, producing a precise written record that serves as primary evidence for later review and auditing; and (2) Acoustic Track, in which the system analyzes the physical properties of the voice by computing the Root Mean Square (RMS) to monitor changes in vocal intensity and applying a three-second sliding window to estimate the user’s speech rate. This dual analysis generates a consolidated data snapshot that integrates dialogue content, volume trajectories, and pacing patterns [27]. During the Evidence Court phase, this record functions as a form of digital evidence grounded in observable behavioral signals. By demonstrating that physical indicators such as a speech rate within 150 to 180 words per minute fall within a typical range, the system provides an objective reference that supports the identification and correction of distorted self-perceptions associated with social anxiety.

4.2 Independent Auditor and Normative Baseline Constraint Implementation

We use an Independent Auditor Agent to score how negative the NPC’s response is in each turn. The auditor is implemented as a secondary GPT-4o instance with low stochasticity (temperature = 0.3) to produce stable ratings across runs [32]. The auditor takes the Whisper-transcribed dialogue as input and outputs an outcome severity score O_{fact} on a 1–10 scale (1 = neutral, 10 = highly negative/rejecting). In our pilot scenarios, we set a conservative safety threshold at $O_{fact} \leq 2$ to reflect that everyday workplace exchanges are typically neutral or only mildly negative.

To enforce this constraint during simulation, the NPC Agent (GPT-4o, $T = 0.7$) is governed by two controls: (1) targeted prompt conditioning that specifies professional and patient conversational norms, and (2) continuous real-time auditing using O_{fact} . After each NPC response, the system queries the auditor. If O_{fact} approaches the threshold, the system blocks escalation and re-prompts the NPC to continue in a neutral-to-mildly-negative range. This design maintains realistic variability while preventing extreme negative outcomes that could overwhelm self-guided exposure. The frontend maps real-time O_{fact} values to the NPC’s multimodal states (e.g., facial expression and tone cues) to support transparency during interaction.

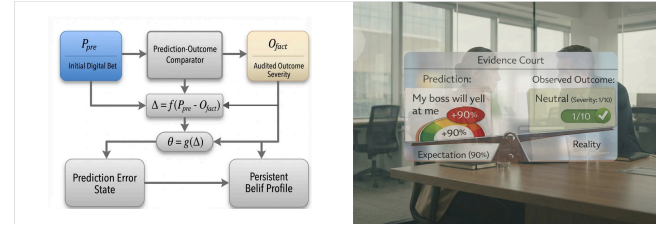


Figure 3: Evidence Court

4.3 Evidence Calibration and Belief Update Implementation

Following each social simulation, the system initiates the rewriting stage of the P–E–R framework by comparing predictions and outcomes. Two backend data dimensions are synchronized: the initial betting probability from the fear formalization stage (P_{pre}) and the objective severity score produced by the Independent Auditor Agent (O_{fact}), which are encapsulated into a unified session object to drive the **Evidence Scale interface** (Figure 3). The Evidence Scale functions as an interactive visualization that externalizes prediction error by computing a visual imbalance angle θ as a function of the discrepancy between P_{pre} and O_{fact} , materialized through CSS-based transformations. To support longitudinal belief calibration, paired prediction-outcome records are persistently stored in a user-specific belief profile and retrieved in subsequent sessions to surface accumulated discrepancies for systematic recalibration.

5 Pilot Study

We conducted a formative pilot study with 8 participants previously involved in user interviews to examine the feasibility and short-term cognitive signals of the P–E–R system. Participants were

assessed using the Liebowitz Social Anxiety Scale (LSAS) [22]. All participants scored above the threshold for mild social anxiety (scores > 30; $M = 61.8$, range = 38–91). Each participant completed three workplace-related social simulation scenarios. The evaluation employed a mixed set of measures, including the System Usability Scale (SUS) [25] for usability, a short engagement scale adapted from prior HCI work, system-logged probabilistic fear estimates (P_{pre} , P_{post}), and auditor-generated outcome severity scores (O_{fact}). Post-session semi-structured interviews were conducted to contextualize quantitative findings [15].

The system demonstrated good usability (SUS: $M = 81.2$, $SD = 7.6$) and sustained engagement ($M = 3.68$, $SD = 0.41$ on a 5-point scale). Participants described the simulations as resembling everyday workplace conversations and reported remaining engaged despite experiencing anxiety. Across scenarios, participants entered with high predicted likelihood of negative outcomes (mean $P_{pre} = 0.76$), while the Independent Auditor consistently rated interactions as neutral or mildly negative (mean $O_{fact} = 1.7/10$). Following Evidence Court reflection, participants reduced their estimates for the same feared outcomes (mean $P_{post} = 0.52$). A Wilcoxon signed-rank test indicated a significant reduction in fear estimates across scenarios ($Z = -2.521$, $p = .016$), consistent with the directional shift observed in mean values. Emotional adaptation features were rated positively, whereas responses to real-time quantitative feedback were mixed, highlighting trade-offs between informational clarity and emotional comfort.

6 Future Work

Future work will focus on strengthening the empirical validation of the proposed framework. While the current study centers on design rationale and mechanism articulation, subsequent research will involve larger and more diverse participant samples to improve generalizability across different levels of social anxiety. Additionally, as the pilot participants were drawn from the same pool as the formative interview cohort, future evaluations will recruit independent samples to reduce potential familiarity bias and demand characteristics. We plan to conduct controlled evaluations that compare the P–E–R system with alternative exposure formats, including unguided role-play, scripted scenario rehearsal, and reflection-only interventions [5]. Longitudinal studies will further examine repeated system use across multiple sessions, incorporating implicit measures or delayed follow-up assessments to distinguish genuine belief updating from socially desirable responding. Subsequent iterations will investigate adaptive adjustment algorithms that gradually increase social ambiguity as users demonstrate reduced baseline anxiety, supporting transfer to the inherently variable conditions of real-world social interaction.

Disclosure of AI Use

Some interface background images and illustrative visual elements in this paper were generated using AI-based image generation tools. These images are used solely for illustrative purposes and do not represent real individuals or real-world environments.

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